



Alberta Public Health Disease Management Guidelines

Plague



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Health and Wellness Promotion Branch

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Alberta Health

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Case Definition

Potential Bioterrorism Agent

Confirmed Case

Clinical illness^(A) with laboratory confirmation of infection:

- Isolation of *Yersinia pestis* from any clinical specimen (e.g., blood, pus, sputum)^(B)

OR

- Seroconversion or significant (i.e. fourfold or greater) rise in serum antibody titre to *Y. pestis* fraction 1 (F1) antigen by EIA or passive hemagglutination inhibition (HI) titre

Probable Case

Clinical illness^(A) with any of the following laboratory evidence:

- Demonstration of elevated serum antibody titre(s) to *Y. pestis* F1 antigen (without documented significant [i.e., fourfold or greater] change) in a patient with no history of plague immunization

OR

- Detection of *Y. pestis* F1 antigen by immunofluorescence

OR

- Detection of *Y. pestis* nucleic acid (e.g., PCR) in an appropriate clinical specimen (e.g., blood, biopsy tissue, CSF)^(C)

OR

- Greater than 1:10 passive HI titre in a single serum sample in a patient with no history of vaccination or previous infection

OR

- Detection of *Y. pestis* antibody by EIA

^(A) Plague is characterized by fever, chills, headache, malaise, prostration and leukocytosis and is manifest in one or more of the following principal forms:

- **Bubonic plague:** regional lymphadenitis
- **Pharyngeal plague:** pharyngitis and cervical lymphadenitis resulting from exposure to larger infectious droplets or ingestion of infected tissues
- **Primary pneumonic plague:** inhalation of infectious droplets
- **Secondary pneumonic plague:** pneumonia, resulting from hematogenous spread in bubonic or septicemic cases
- **Septicemic plague:** septicemia with or without an evident bubo

^(B) Refer to the [Public Health Laboratory \(ProvLab\) Guide to Services](#) for current specimen collection and submission information.

^(C) Refer to the [National Microbiology Laboratory Guide to Services](#) for current specimen collection and submission information.

Reporting Requirements

Physicians, Health Practitioners and Others

Physicians, health practitioners and others shall notify the Medical Officer of Health (MOH) (or designate) of the zone, of all confirmed and probable cases in the prescribed form by the Fastest Means Possible (FMP).

Laboratories

All laboratories shall report all positive laboratory results:

- by FMP to the MOH (or designate) of the zone, and
- by mail, fax or electronic transfer within 48 hours (two business days) to the Chief Medical Officer of Health (CMOH) (or designate).

Alberta Health Services and First Nations and Inuit Health Branch

- The MOH (or designate) of the zone where the case currently resides shall:
 - notify the CMOH (or designate) by FMP of all confirmed and probable cases, and
 - forward the initial Notifiable Disease Report (NDR) of all confirmed and probable cases to the CMOH (or designate) within one week of notification and the final NDR (amendment) within two weeks of notification.
- For out-of-province and out-of-country reports, the following information should be forwarded to the CMOH (or designate) by FMP:
 - name,
 - date of birth,
 - out-of-province health care number,
 - out-of-province address and phone number,
 - positive laboratory report, and
 - other relevant clinical/epidemiological information.

Appendix 1: Revision History

Revision Date	Document Section	Description of Revision
November 2021	General	• Updated Template • Updated web links